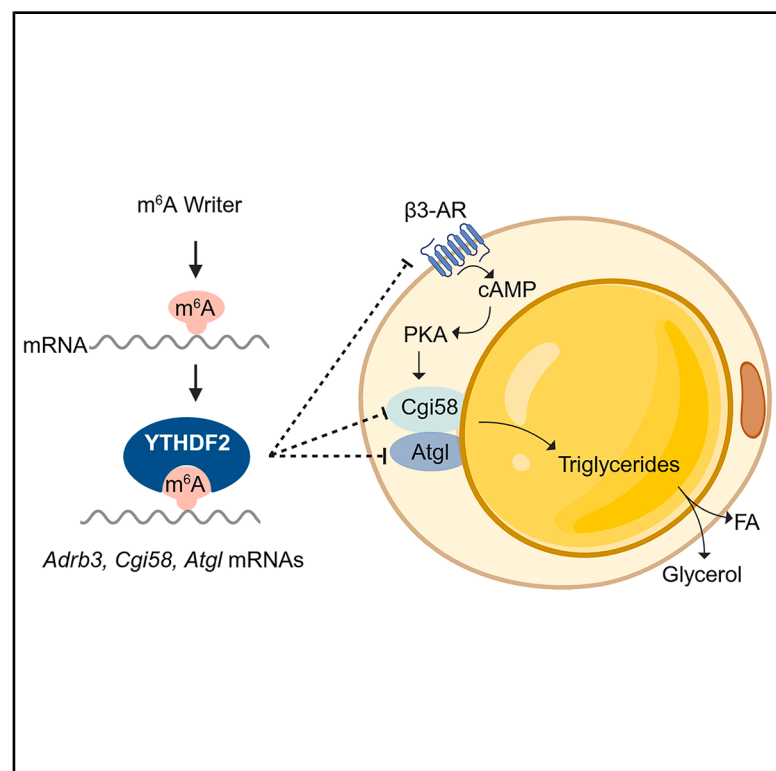


Adipose m⁶A reader YTHDF2 promotes obesity, insulin resistance, and liver steatosis by suppressing β -adrenergic signaling and lipolysis

Graphical abstract



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In brief

Zhou et al. identify YTHDF2 as an epitranscriptomic regulator of adipose expansion and dysfunction. YTHDF2 binds to m⁶A-modified transcripts encoding the β 3-receptor, ATGL, and CGI-58, promoting their degradation and, thereby, inhibiting β -adrenergic signaling and lipolysis. YTHDF2 emerges as a long-sought mediator underlying adipose catecholamine resistance and impaired lipolysis in obesity.

Highlights

- YTHDF2 promotes the degradation of m⁶A-modified *Adrb3*, *Atgl*, and *Cgi-58* mRNAs
- YTHDF2 inhibits both adipose β -adrenergic signaling and lipolysis
- YTHDF2 is dispensable for brown adipose tissue development
- Deletion of adipose *Ythdf2* alleviates diet-induced obesity and metabolic disorders



Article

Adipose m⁶A reader YTHDF2 promotes obesity, insulin resistance, and liver steatosis by suppressing β -adrenergic signaling and lipolysis

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SUMMARY

RNA N⁶-methyladenosine (m⁶A) modification induces catecholamine resistance and lipolysis inhibition in white adipose tissue (WAT), contributing to obesity pathogenesis; however, the responsible m⁶A readers remain elusive. Here, we identify YTHDF2 as a key m⁶A reader governing both β -adrenergic signaling and lipolytic machinery. YTHDF2 binds to m⁶A-marked mRNAs encoding β 3-adrenergic receptor (*Adrb3*), adipose triacylglycerol lipase (*Atgl*), and comparative gene identification-58 (*Cgi-58*), promoting their degradation and thereby suppressing β -adrenergic signaling and lipolysis. Deletion of adipose *Ythdf2* enhances lipolysis *in vivo*, in WAT explants *ex vivo*, and in cultured adipocytes. Conversely, YTHDF2 overexpression suppresses adipocyte lipolysis. High-fat diet feeding upregulates adipose YTHDF2 and increases its binding to *Adrb3*, *Atgl*, and *Cgi-58* mRNAs. Adipocyte-specific deletion of *Ythdf2* protects against diet-induced obesity, insulin resistance, and liver steatosis. Moreover, deletion of adipose *Mettl14*—but not *Ythdf2*—disrupts brown adipose tissue development. These results unveil an adipose-intrinsic METTL3/METTL14/m⁶A/YTHDF2 pathway that drives catecholamine resistance and lipolysis suppression in obesity.

INTRODUCTION

Lipolysis mobilizes free fatty acids (FFAs) from white adipose tissue (WAT) as metabolic fuel to support life in response to fasting, cold exposure, exercise, and immune reactions to pathogens. Lipolysis-released FFAs also serve as the structural component of phospholipids that are required for the synthesis and remodeling of biological membranes. Furthermore, lipolysis-derived bioactive lipids orchestrate numerous critical physiological processes, safeguard body homeostasis, and function as autocrine, paracrine, and/or endocrine mediators. Adipose triacylglycerol lipase (ATGL), together with cofactor CGI-58, catalyzes the initial step of lipolysis by hydrolyzing triacylglycerol (TAG) stored in lipid droplets to release FFAs^{1–4}. Sympathetic nerves potently stimulate lipolysis through a β -adrenergic receptor/protein kinase A (PKA)/cyclic AMP (cAMP) pathway in adipocytes.^{5–7} Notably, adipose ATGL and CGI-58 are downregulated and lipolysis is inhibited in obesity.^{8–13} Moreover, adipose sympathetic nerve transmis-

sions and β -adrenergic signaling are also impaired in obesity, resulting in catecholamine resistance.^{10,14–16} However, the mechanisms underlying obesity-associated catecholamine resistance and lipolysis suppression remain elusive.

N⁶-methyladenosine (m⁶A) modification and m⁶A-dependent RNA-binding proteins have emerged as key regulators of cell proliferation, differentiation, and function.^{17,18} METTL3 and METTL14 are the catalytic and structural subunits, respectively, of the RNA methyltransferase complex, also known as the m⁶A writer, which installs m⁶A on coding and noncoding RNAs.^{19–22} FTO and ALKBH5 act as m⁶A erasers to demethylate RNA, thereby influencing the dynamic landscape of m⁶A modifications.^{23–25} A growing number of m⁶A-dependent RNA-binding proteins, called m⁶A readers, have been identified and shown to regulate RNA metabolism/fate, such as pre-RNA splicing, RNA decay, mRNA nuclear export, and/or translation. These include members of the YTH family—YTHDF1–3 and YTHDC1–2^{17–19,26–31}. Interestingly, high-fat diet (HFD)-induced obesity is associated with upregulation of METTL3 and METTL14 in WAT



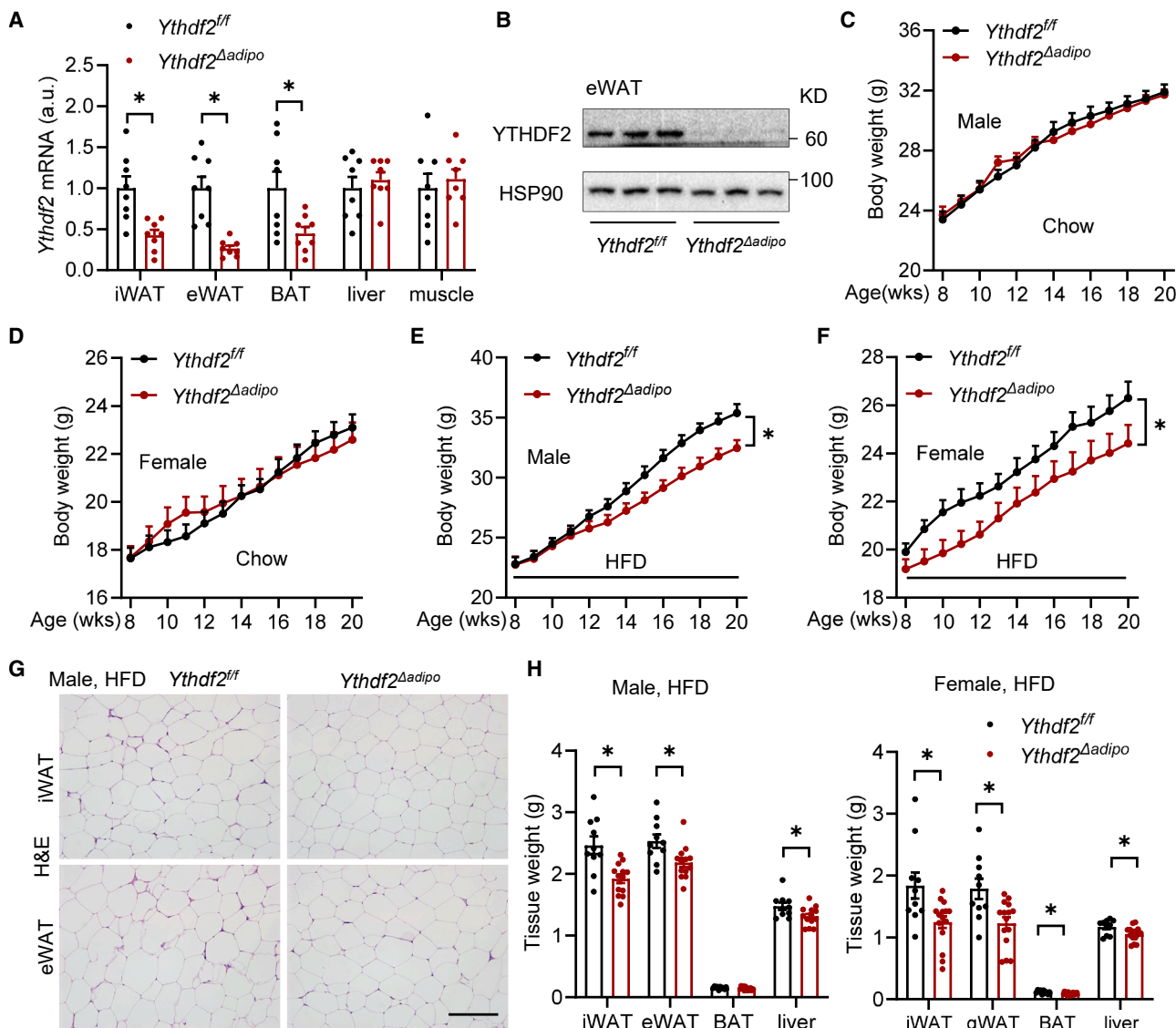


Figure 1. Adipocyte-specific deletion of *Ythdf2* protects against HFD-induced obesity

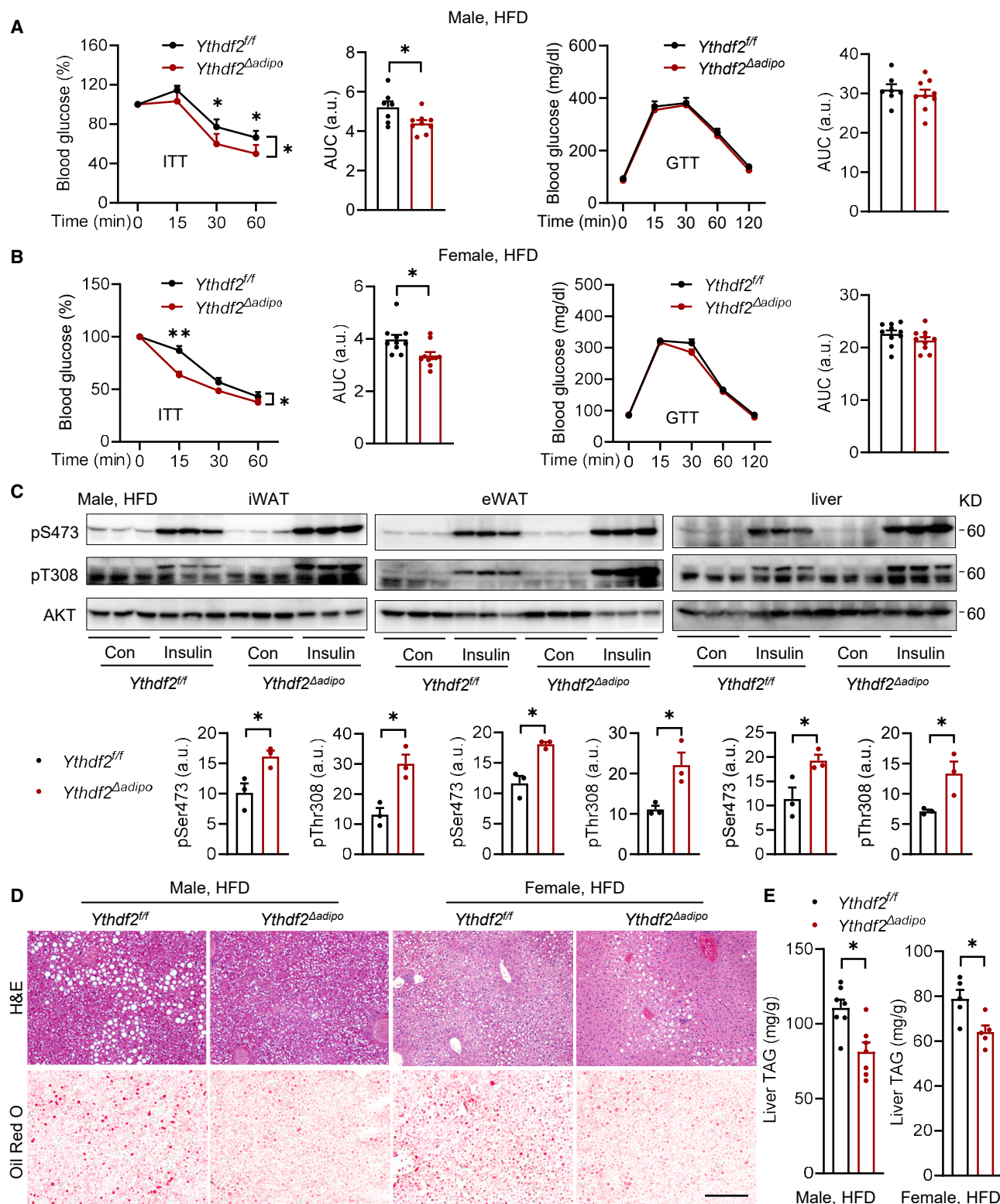
(A–D) Mice were fed a normal chow diet. (A) *Ythdf2* mRNA levels were measured by qPCR and normalized to 36B4 expression at 8 weeks of age ($n = 8$ mice per group). (B) eWAT extracts were immunoblotted with anti-YTHDF2 and anti-HSP90 antibodies. (C) and (D) Body weight. Male *Ythdf2*^{fl/fl}: $n = 6$, *Ythdf2*^{Δadipo}: $n = 7$; female *Ythdf2*^{fl/fl}: $n = 9$, and *Ythdf2*^{Δadipo}: $n = 9$.

(E–H) Mice were fed HFD. (E) and (F) Body weight. Male *Ythdf2*^{fl/fl}: $n = 10$, *Ythdf2*^{Δadipo}: $n = 13$; female *Ythdf2*^{fl/fl}: $n = 10$, and *Ythdf2*^{Δadipo}: $n = 13$. (G) H&E staining of WAT sections (HFD for 16 weeks). Scale bars, 200 μ m. (H) Tissue weight on HFD for 20 weeks. Male *Ythdf2*^{fl/fl}: $n = 10$, *Ythdf2*^{Δadipo}: $n = 13$; female *Ythdf2*^{fl/fl}: $n = 10$, and *Ythdf2*^{Δadipo}: $n = 15$.

Data are presented as mean $\pm$ SEM. * $p < 0.05$, two-sided unpaired *t* test for (A) and (H), and two-way ANOVA test for (C–F). See also Figure S1.

and the liver, two key metabolic organs,^{10,32} suggesting that the m⁶A pathway may play roles in regulating metabolic homeostasis. Supporting this notion, adipocyte-specific deletion of *Mettl14* (*Mettl14*^{Δadipo}) attenuates HFD-induced WAT expansion, insulin resistance, and metabolic dysfunction-associated steatotic liver disease (MASLD).¹⁰ Similarly, deletion of adipose *Mettl3* also alleviates HFD-induced obesity.³³ Adipose METTL14 promotes WAT expansion at least in part by inducing catecholamine resistance and suppressing lipolysis.¹⁰ Deletion of hepatic *Mettl14* suppresses liver glucose production through downregulation of

glucose-6-phosphatase by an epitranscriptomic mechanism.³² In this work, we aimed to identify m⁶A readers that mediate the metabolic function of the METTL14/METTL3 complex, focusing on YTHDF2, as genetic variants of YTHDF2 have been associated with type 2 diabetes in humans.³⁴ Global deletion of *Ythdf2* in mice results in embryonic lethality, indicating that YTHDF2 is required for development and survival.³⁵ Beyond recognizing m⁶A-marked mRNAs, YTHDF2 also interacts with the CCR4–NOT deadenylase complex to promote deadenylation and degradation of its targets.³⁶ In addition, YTHDF2 recruits UPF1 and



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RNase P/MRP, which mediates mRNA decapping and degradation, respectively.^{37,38} YTHDF2 also facilitates the translocation of m⁶A-marked RNAs from the translational pool to decay sites (e.g., P-bodies), thereby promoting RNA turnover.³¹ Here, we investigated the function of adipose YTHDF2 by generating and characterizing adipocyte-specific *Ythdf2* knockout mice (*Ythdf2*^{Δadipo}), with a focus on the development and metabolic function of WAT and brown adipose tissue (BAT). We examined in depth the YTHDF2-regulated molecular events orchestrating β-adrenergic signaling and lipolysis. Our findings unveil previously unrecognized m⁶A/YTHDF2/β3 adrenergic receptor and m⁶A/YTHDF2/ATGL/CGI-58 pathways. These pathways play critical roles in promoting catecholamine resistance and suppressing lipolysis, thereby contributing to excessive WAT expansion and metabolic disorders in obesity.

RESULTS

Adipocyte-specific deletion of *Ythdf2* alleviates HFD-induced obesity

To explore adipose YTHDF2, we generated *Ythdf2*^{Δadipo} mice by crossing *Ythdf2*^{fl/fl} mice with *adiponectin-cre* drivers. The start codon-containing exon 1 was deleted in *Ythdf2*^{Δadipo} mice (Figure S1A). As expected, *Ythdf2* mRNA levels were significantly lower in the inguinal WAT (iWAT), epididymal WAT (eWAT), and BAT but not liver and skeletal muscle, of *Ythdf2*^{Δadipo} mice relative to *Ythdf2*^{fl/fl} mice (Figure 1A). YTHDF2 protein was readily detected in eWAT in *Ythdf2*^{fl/fl} but not *Ythdf2*^{Δadipo} mice (Figure 1B). Body weight was normal in both male and female *Ythdf2*^{Δadipo} mice under a standard chow diet (Figures 1C and 1D). Fat content and weights of iWAT, eWAT (males), gonadal WAT (gWAT, females), BAT, and liver were similar between *Ythdf2*^{Δadipo} and *Ythdf2*^{fl/fl} mice (Figures S1B and S1C). Next, *Ythdf2*^{Δadipo} and control mice were fed an HFD to induce obesity. Both male and female *Ythdf2*^{Δadipo} mice gained significantly less body weight relative to sex-matched *Ythdf2*^{fl/fl} littermates (Figures 1E and 1F). Their white adipocytes were smaller (Figures 1G and S1D). The weights of iWAT (both males and females), eWAT (males), and gWAT (females) were significantly lower in *Ythdf2*^{Δadipo} than in *Ythdf2*^{fl/fl} littermates (Figure 1H). In contrast, lean mass was comparable between *Ythdf2*^{Δadipo} and *Ythdf2*^{fl/fl} mice (Figure S1E). These results suggest that adipose YTHDF2 promotes diet-induced WAT expansion in wild-type control mice.

Ythdf2^{Δadipo} mice are resistant to HFD-induced insulin resistance and liver steatosis

To determine whether adipose YTHDF2 regulates systemic metabolism, we performed insulin tolerance tests (ITTs) and

glucose tolerance tests (GTTs) on *Ythdf2*^{Δadipo} and control mice under HFD for 12 weeks. YTHDF2 deficiency increased the ability of insulin to lower blood glucose in both male and female *Ythdf2*^{Δadipo} mice compared to *Ythdf2*^{fl/fl} littermates; however, GTT was comparable between *Ythdf2*^{Δadipo} and control mice (Figures 2A and 2B). To examine insulin signaling, *Ythdf2*^{Δadipo} mice (on HFD for 16 weeks) were fasted overnight and injected with insulin for 5 min. Insulin-stimulated phosphorylation of AKT (pSer473 and pThr308) in iWAT, eWAT, and livers was significantly higher in *Ythdf2*^{Δadipo} than in *Ythdf2*^{fl/fl} mice (Figure 2C). Liver weight was significantly lower, due to decreased lipid levels, in *Ythdf2*^{Δadipo} mice relative to sex-matched *Ythdf2*^{fl/fl} mice (Figure 1H). H&E and Oil red O (a neutral lipid dye) staining of liver sections confirmed that lipid droplets were smaller and less abundant in *Ythdf2*^{Δadipo} mice on HFD for 16 weeks (Figure 2D). Liver TAG levels were significantly lower in both male and female *Ythdf2*^{Δadipo} mice compared to controls (Figure 2E). Consistently, expression of the genes involved in lipid uptake and lipogenesis (*Cd36* and *Dgat1*) was reduced while fatty acid β oxidation genes (*Ppara* and *Pgc1α*) were upregulated in the liver of *Ythdf2*^{Δadipo} mice (Figure S2A). Notably, prior to the onset of body weight divergence, ITT, GTT, and liver lipid levels were indistinguishable between *Ythdf2*^{Δadipo} and *Ythdf2*^{fl/fl} mice (Figures S2B and S2C), suggesting that the fat mass difference between *Ythdf2*^{Δadipo} and *Ythdf2*^{fl/fl} mice primarily contributes to their distinct metabolic phenotypes. Collectively, these results demonstrate that adipose YTHDF2 promotes diet-induced obesity, insulin resistance, and liver steatosis.

YTHDF2 suppresses lipolysis in adipocytes

We next set out to investigate how YTHDF2 promotes WAT expansion. Given that METTL14, a m⁶A writer, inhibits lipolysis,¹⁰ we reasoned that YTHDF2 (a m⁶A reader) might act downstream of METTL14 to inhibit lipolysis, thereby increasing TAG accumulation and WAT expansion. *Ythdf2*^{Δadipo} and *Ythdf2*^{fl/fl} littermates (on HFD for 8 weeks) were injected with isoproterenol (Iso), β-adrenergic agonist, to increase their lipolysis. Iso increased plasma FFA and glycerol, two lipolysis products, to a significantly higher level in *Ythdf2*^{Δadipo} than in *Ythdf2*^{fl/fl} mice (Figure 3A). To confirm these findings, iWAT and eWAT were isolated from *Ythdf2*^{Δadipo} and *Ythdf2*^{fl/fl} mice (on HFD for 8 weeks) and used for *ex vivo* lipolysis assays (Figure 3B). Iso-stimulated lipolysis was significantly higher in *Ythdf2*^{Δadipo} iWAT and eWAT explants than in controls, albeit baseline lipolysis was comparable between the two groups (Figures 3C, 3D, and S3A). Iso-stimulated lipolysis was also significantly increased

Figure 2. *Ythdf2*^{Δadipo} mice are resistant to HFD-induced insulin resistance and liver steatosis

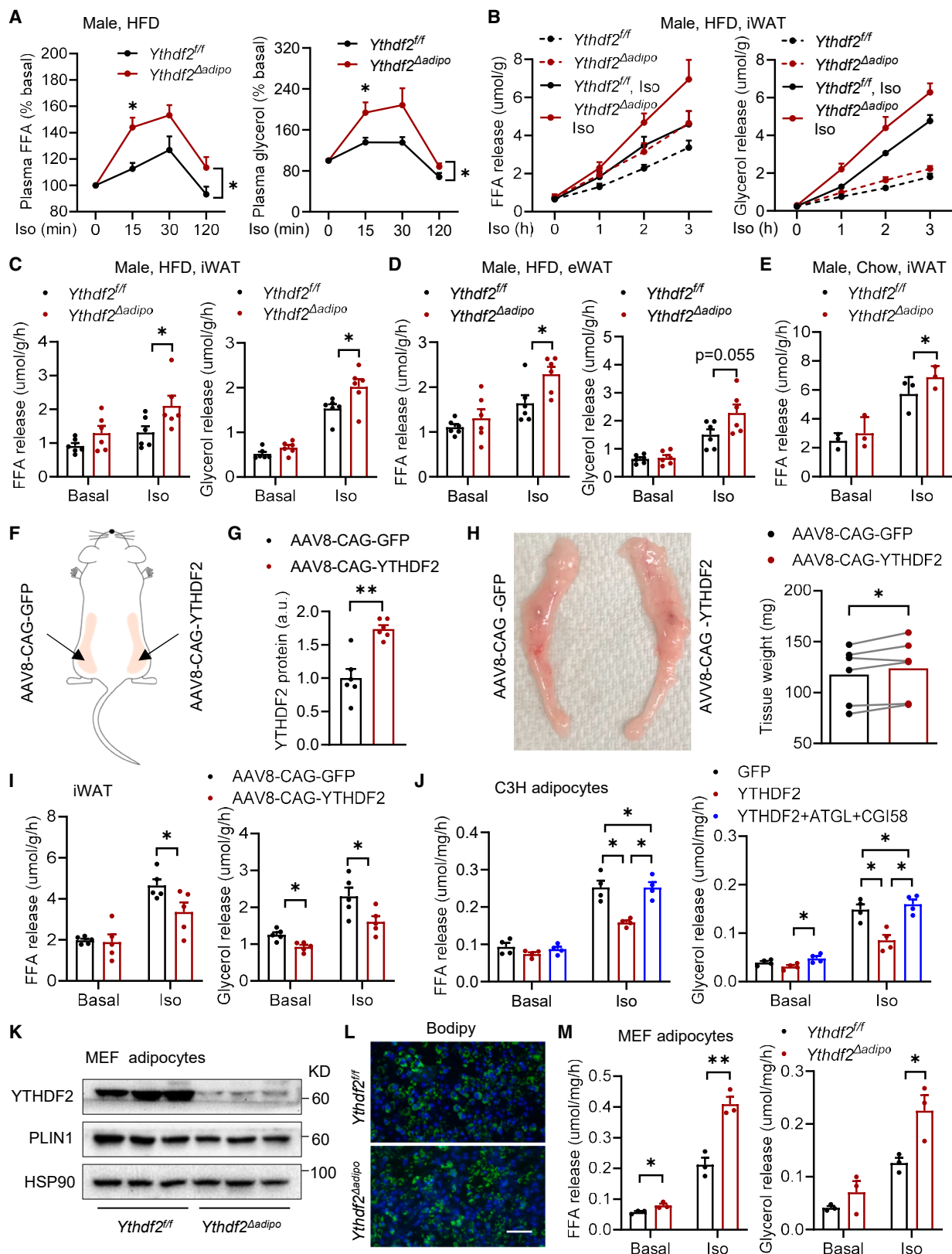
(A and B) ITT and GTT on HFD for 12 weeks. Male *Ythdf2*^{fl/fl}: *n* = 7, *Ythdf2*^{Δadipo}: *n* = 9; female *Ythdf2*^{fl/fl}: *n* = 10, and *Ythdf2*^{Δadipo}: *n* = 10. AUC, area under the curve. a.u., arbitrary units.

(C) Male mice were fed HFD for 16 weeks, fasted overnight, and injected with insulin (0.6 U/kg) for 5 min. WAT and liver extracts were immunoblotted with anti-phospho-AKT (pThr308 and pSer473). AKT phosphorylation was normalized to total AKT levels (*n* = 3 mice per group).

(D) H&E and Oil red O staining of liver sections (HFD for 16 weeks). Scale bars, 200 μm.

(E) Liver TAG levels on HFD for 16 weeks (normalized to liver weight).

Data are presented as mean ± SEM. **p* < 0.05 and ***p* < 0.01, two-way ANOVA (A and B), two-sided unpaired *t*-tests (A–C) and (E). See also Figure S2.



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in iWAT explants isolated from chow-fed *Ythdf2^{Δadipo}* mice (Figures 3E and S3B).

To test whether YTHDF2 overexpression exerts opposite effects, an AAV8-CAG-YTHDF2 vector was injected into the iWAT depot of C57BL/6J mice, while AAV8-CAG-GFP vector was injected into the contralateral iWAT depot as a control (Figure 3F). YTHDF2 levels were increased in AAV8-CAG-YTHDF2-transduced iWAT depots compared to controls (Figures 3G and S3C). Importantly, YTHDF2-overexpressing iWAT depots were significantly enlarged 4 weeks after AAV injection compared to controls (Figure 3H). Concomitantly, Iso-stimulated lipolysis was significantly reduced in AAV8-CAG-YTHDF2-transduced iWAT depots (Figures 3I and S3D). To test if YTHDF2 directly suppresses lipolysis, C3H preadipocytes were transfected with YTHDF2 or GFP (control) plasmids, differentiated into adipocytes, and then stimulated with Iso. Although YTHDF2 was only modestly overexpressed (Figure S3E), it potently suppressed Iso-stimulated lipolysis under these conditions (Figures 3J and S3F). To further validate these findings, mouse embryonic fibroblasts (MEFs) were prepared from *Ythdf2^{Δadipo}* and *Ythdf2^{fl/fl}* embryos and differentiated into adipocytes *in vitro*. YTHDF2 protein was detected in *Ythdf2^{fl/fl}* but not *Ythdf2^{Δadipo}* adipocytes as expected (Figure 3K). Importantly, Iso-stimulated lipolysis was significantly higher in *Ythdf2^{Δadipo}* adipocytes compared to *Ythdf2^{fl/fl}* controls, despite normal adipogenesis in *Ythdf2^{Δadipo}* MEFs (Figures 3L, 3M, and S3G). These results uncover YTHDF2 as a previously unrecognized epitranscriptomic suppressor of lipolysis.

YTHDF2 inhibits the biosynthesis of the components of the β -adrenergic pathway and the lipolysis machine

To identify YTHDF2-targeted molecules, we harvested iWAT and eWAT from *Ythdf2^{Δadipo}* and *Ythdf2^{fl/fl}* mice fed an HFD for 16 weeks and measured the expression of ATGL, CGI-58, hormone-sensitive lipase (HSL), and *Adrb3* (β 3-adrenergic receptor). ATGL, CGI-58, and *Adrb3*, but not HSL, in both iWAT and eWAT, were significantly upregulated in *Ythdf2^{Δadipo}* mice compared to *Ythdf2^{fl/fl}* controls (Figure 4A). To examine β -adrenergic signaling, iWAT and eWAT explants were stimulated with Iso, and tissue extracts were immunoblotted with phospho-antibodies to PKA substrates (pan-pPKA) and HSL (pSer563 and

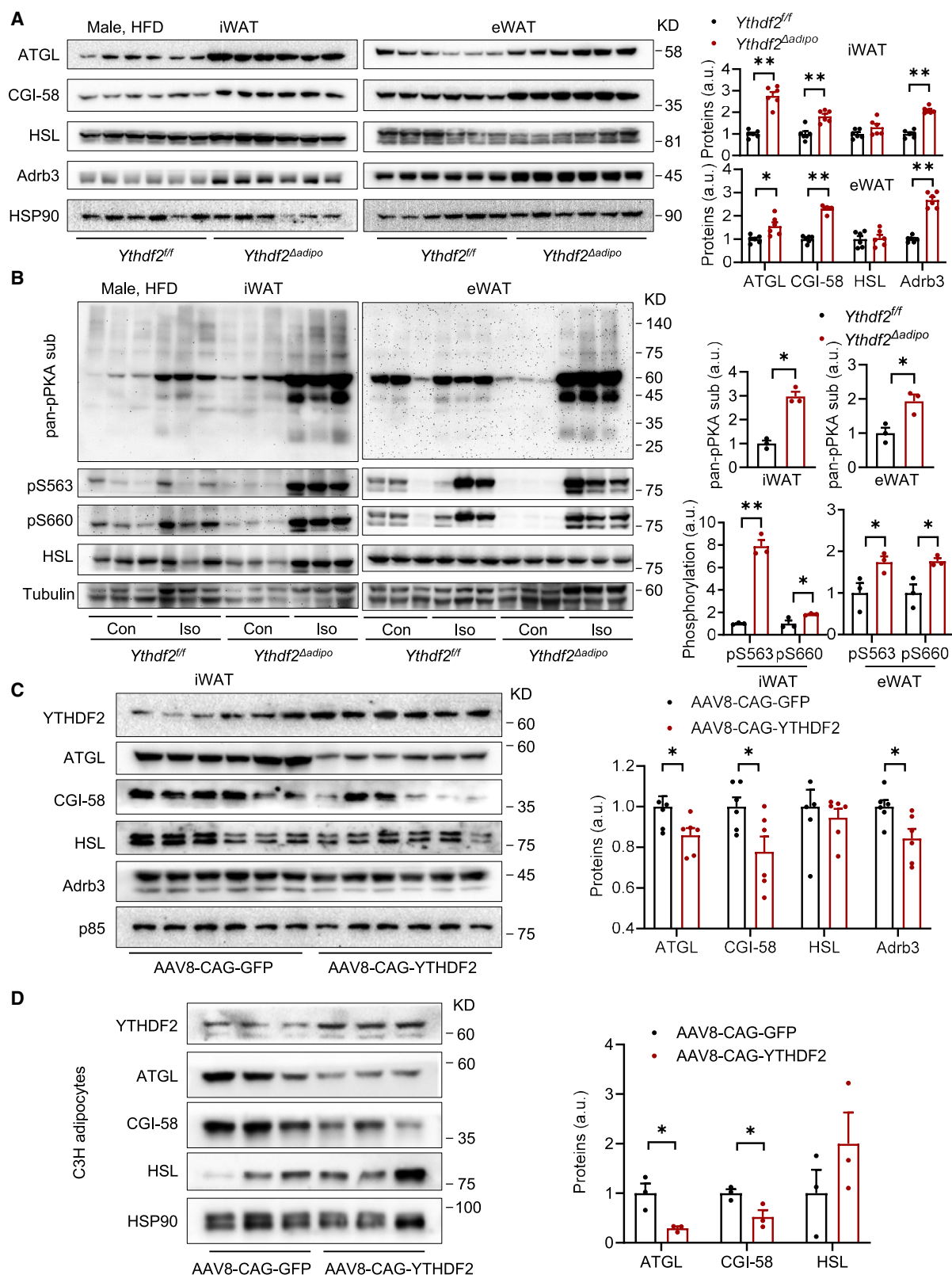
pSer660). Iso-stimulated phosphorylation of PKA substrates, including HSL (pSer563 and pSer660), was significantly elevated in WAT explants from *Ythdf2^{Δadipo}* mice compared to *Ythdf2^{fl/fl}* controls (Figure 4B). To test if the overexpression of YTHDF2 exerts the opposite effects, an AAV8-CAG-YTHDF2 vector was injected into the iWAT depot of C57BL/6J mice, while an AAV8-CAG-GFP vector was injected into the contralateral iWAT depot as a control. YTHDF2 overexpression considerably decreased *Adrb3*, ATGL, and CGI-58 levels in AAV8-CAG-YTHDF2-transduced iWAT depots (Figure 4C). To further confirm that YTHDF2 cell autonomously downregulates ATGL and CGI-58, C3H adipocytes were transduced with AAV8-CAG-YTHDF2 or AAV8-CAG-GFP vector. YTHDF2 significantly decreased ATGL and CGI-58 levels (Figure 4D). Moreover, restoration of ATGL and CGI-58 expression in YTHDF2-overexpressing adipocytes reversed lipolysis inhibition (Figure 3J). These results unravel previously unrecognized YTHDF2/ β -adrenergic signaling/lipolysis and YTHDF2/ATGL/CGI-58/lipolysis loops.

YTHDF2 binds m⁶A-modified *Adrb3*, *Atgl*, and *Cgi-58* mRNAs to induce mRNA degradation

We postulated that YTHDF2 promotes the degradation of its bound *Adrb3*, *Atgl*, and *Cgi-58* mRNAs, thereby contributing to catecholamine resistance and lipolysis suppression. We aimed to detect YTHDF2-bound transcripts using YTHDF2-linked RNA immunoprecipitation (RIP) assays. YTHDF2-bound mRNAs were immunoprecipitated from the eWAT of *Ythdf2^{fl/fl}* and *Ythdf2^{Δadipo}* (a negative control) mice using an anti-YTHDF2 antibody and quantified by qPCR. *Adrb3*, *Atgl*, and *Cgi-58* mRNA levels were significantly lower in *Ythdf2^{Δadipo}* than in *Ythdf2^{fl/fl}* eWAT (Figure 5A). To confirm these findings, C3H adipocytes were transduced with AAV8-CAG-FLAG-YTHDF2 or AAV8-CAG-GFP vector (control). FLAG-YTHDF2-bound mRNAs were immunoprecipitated from cell lysates with FLAG beads and quantified by qPCR. *Adrb3*, *Atgl*, and *Cgi-58* mRNA levels were significantly higher in AAV8-CAG-FLAG-YTHDF2 transduced adipocytes (Figure 5B). m⁶A modification of *Adrb3*, *Atgl*, and *Cgi-58* mRNAs was impaired in *Mettl14^{Δadipo}* mice,¹⁰ allowing us to test whether this impairment disrupts the interaction between YTHDF2 and these transcripts. Indeed, YTHDF2-bound *Adrb3*, *Atgl*, and *Cgi-58* mRNA levels were significantly lower

Figure 3. YTHDF2 suppresses lipolysis in adipocytes to increase WAT growth

(A) Male mice were fed HFD for 8 weeks, fasted overnight, and injected with isoproterenol (Iso, 10 mg/kg body weight). Plasma FFA and glycerol levels were measured and normalized to initial values. *Ythdf2^{fl/fl}*: *n* = 7 and *Ythdf2^{Δadipo}*: *n* = 8.
(B–D) Male mice were fed HFD for 12 weeks. WAT was isolated and stimulated with 1 μ M isoproterenol for 3 h *ex vivo*. FFA and glycerol releases were measured and normalized to WAT weight (*n* = 6 mice per group).
(E) iWAT was isolated from male mice at 16 weeks of age on chow diet. iWAT was stimulated with 1 μ M isoproterenol for 3 h. FFA release from iWAT was measured *ex vivo* and normalized to WAT weight (*n* = 3 mice per group).
(F–I) C57BL/6J males were bilaterally injected with AAV8-CAG-GFP and AAV8-CAG-YTHDF2 vectors, respectively (F). AAV-transduced iWAT was harvested 4 weeks later. (G) iWAT extracts were immunoblotted with anti-YTHDF2 antibody, and YTHDF2 levels were normalized to p85 levels (loading control) (*n* = 6 mice per group). (H) Images and weights of AAV-transduced iWAT. (I) iWAT was stimulated with 1 μ M isoproterenol for 3 h, and FFA and glycerol releases were measured and normalized to WAT weight *ex vivo* (*n* = 5 mice per group).
(J) C3H preadipocytes were transfected with YTHDF2, ATGL, and CGI-58 plasmids and differentiated into adipocytes. Adipocytes were stimulated with 1 μ M isoproterenol for 3 h, and FFA and glycerol releases were measured and normalized to protein levels (*n* = 4 per group).
(K–M) MEF cells were prepared from *Ythdf2^{fl/fl}* and *Ythdf2^{Δadipo}* embryos and differentiated into adipocytes (*n* = 3 per group). (K) Adipocyte extracts were immunoblotted with the indicated antibodies. (L) Adipocytes were stained with lipid droplet dye BODIPY. Scale bars, 125 μ m. (M) Adipocytes were stimulated with 1 μ M isoproterenol for 3 h, and FFA and glycerol releases were measured and normalized to protein levels (*n* = 3 per group).
Data are presented as mean $\pm$ SEM. **p* < 0.05 and ***p* < 0.01, two-way ANOVA (A) and (B), two-sided unpaired t test (C–E), (G), (I), (J), and (M), and two-sided paired t test with Sidák's multiple-comparison test (H). See also Figure S3.



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in *Mettl14^{Δadipo}* than in *Mettl14^{fl/fl}* mice (Figure 5C), suggesting that YTHDF2 binding to these transcripts is m⁶A-dependent.

In iWAT and eWAT, *Adrb3*, *Atgl*, and *Cgi-58* mRNA levels were significantly higher in *Ythdf2^{Δadipo}* than in *Ythdf2^{fl/fl}* mice fed an HFD for 16 weeks (Figures 5D and 5E). Conversely, YTHDF2 overexpression significantly decreased *Adrb3*, *Atgl*, and *Cgi-58* mRNA levels in iWAT *in vivo* (Figure 5F). In C3H adipocyte culture, YTHDF2 overexpression also reduced *Adrb3*, *Atgl*, and *Cgi-58* expressions (Figure 5G). YTHDF2 is known to promote mRNA decay, prompting us to assess the degradation of *Adrb3*, *Atgl*, and *Cgi-58* mRNAs using transcription inhibitor actinomycin D. In C3H adipocytes, YTHDF2 overexpression markedly accelerated the degradation of these transcripts, resulting in reduced mRNA half-lives (t_{1/2}) (Figure 5H). To examine the role of m⁶A, C3H preadipocytes were transfected with YTHDF2 or GFP (control) plasmids, differentiated into adipocytes, and then treated with STM2457, a METTL3 inhibitor that blocks RNA m⁶A methylation.¹⁰ STM2457 largely abrogated the ability of YTHDF2 to downregulate *ATGL* and *CGI-58* (Figures 5I, S4A, and S4B). In line with these findings, we previously reported that the METTL14-containing m⁶A writer installs m⁶A modifications on *Adrb3*, *Atgl*, and *Cgi-58* transcripts.¹⁰ Taken together, our results unveil a YTHDF2/m⁶A-mRNA (*Adrb3*)/β-adrenergic signaling pathway and a YTHDF2/m⁶A-mRNA (*ATGL* and *CGI-58*)/lipolysis machinery axis that collectively coordinates adipose metabolic regulation.

We further observed that genes encoding lipid droplet proteins, such as *Plin1* and *Cidea*, were downregulated in *Ythdf2^{Δadipo}* mice, although YTHDF2 did not directly bind to these mRNAs (Figures S4C and S4D), suggesting an indirect regulatory mechanism. In addition, expression of several inflammatory markers, including F4/80 and tumor necrosis factor α (TNFα), was reduced in WAT of *Ythdf2^{Δadipo}* mice fed an HFD (Figure S4D). Expression of fibrotic genes, such as α-smooth muscle actin (αSMA), was also reduced in the eWAT of *Ythdf2^{Δadipo}* mice (Figure S4D). These findings suggest that in addition to suppressing β-adrenergic signaling and lipolysis, YTHDF2 deficiency promotes metabolically beneficial adipose remodeling. To further explore the broader landscape of YTHDF2 targets, we reanalyzed publicly available m⁶A-marked RNA datasets (GSE250137) and identified 6024 m⁶A-modified transcripts in WAT from wild-type but not METTL14-deficient mice, including *Adrb3*, *Atgl*, *Hsl*, and *Cgi-58* mRNAs (Figure S5A). These m⁶A-marked transcripts are involved in diverse metabolic processes, such as lipid metabolism, transport, localization, and phospholipid biosynthesis (Figure S5B). We propose that YTHDF2 may regulate these pathways by tar-

geting additional m⁶A-modified transcripts beyond *Adrb3*, *Atgl*, and *Cgi-58*.

Adrb3, *Atgl*, and *Cgi-58* mRNA m⁶A methylations are elevated in WAT under diet-induced obesity, likely due to the upregulation of METTL14 and METTL3.¹⁰ This observation prompted us to test whether YTHDF2 is involved in mediating the downregulation of these m⁶A-modified transcripts, contributing to catecholamine resistance and suppressed lipolysis in obesity. Indeed, YTHDF2 protein and mRNA levels were significantly increased in the iWAT and eWAT of HFD-fed C57BL/6J mice compared to chow-fed controls (Figures 5J and 5K). Furthermore, YTHDF2-bound *Adrb3*, *Atgl*, and *Cgi-58* mRNA levels were significantly elevated in the iWAT and eWAT of HFD-fed mice (Figure 5L). To explore the mechanism underlying YTHDF2 upregulation, C3H cells were differentiated into adipocytes and treated with palmitic acids (PA), TNFα, or insulin. Among these, TNFα significantly increased YTHDF2 levels (Figure S5C), suggesting that proinflammatory cytokines may contribute to obesity-associated upregulation of adipose YTHDF2. Collectively, our data suggest that obesogenic factors stimulate adipose METTL3/METTL14/YTHDF2 epitranscriptomic pathway, thereby inducing catecholamine resistance and pathogenic lipolysis suppression.

Ambient temperature influences YTHDF2's action in WAT

Given the critical role of adipose lipolysis in thermogenic responses to cold exposure,⁵ we investigated whether chronic cold acclimation alters the phenotypes of *Ythdf2^{Δadipo}* mice. *Ythdf2^{Δadipo}* and *Ythdf2^{fl/fl}* male mice were housed at 16°C and fed HFD for 12 weeks. *Ythdf2^{Δadipo}* mice gained markedly less body weight compared to *Ythdf2^{fl/fl}* controls (Figure 6A). Both iWAT and eWAT weights were significantly lower in *Ythdf2^{Δadipo}* than in *Ythdf2^{fl/fl}* mice (Figure 6B). In ITT, blood glucose levels were significantly lower in *Ythdf2^{Δadipo}* mice after insulin injection (Figure 6C), whereas glucose tolerance remained relatively normal (Figure 6D). *Ythdf2^{Δadipo}* mice were resistant to HFD-induced liver steatosis, as evidenced by reduced hepatic lipid droplet accumulation and lower TAG levels (Figures 6E and 6F). Notably, the difference in body weight between *Ythdf2^{fl/fl}* and *Ythdf2^{Δadipo}* mice on HFD was significantly greater at an ambient temperature of 16°C (5 g) compared to 22°C (2.9 g) (Figure 6G). Upon transitioning from 22°C to 16°C, *Ythdf2^{Δadipo}* mice lost 3.7 g, whereas *Ythdf2^{fl/fl}* mice lost only 1.6 g (Figure 6G). Similarly, the differences in iWAT and eWAT weights between the two genotypes were more pronounced at 16°C than at 22°C (Figure 6H). However, food intake was comparable

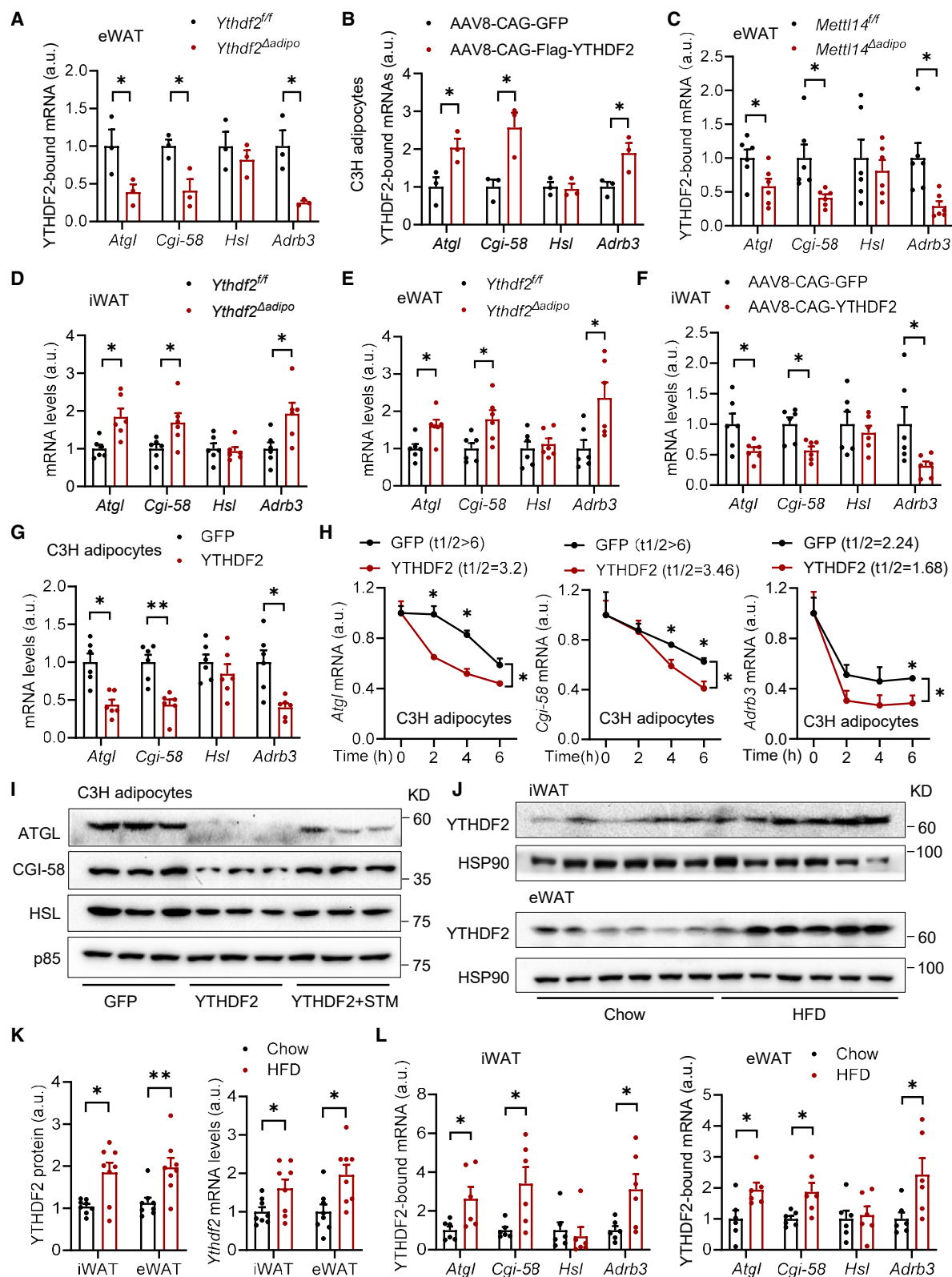
Figure 4. YTHDF2 inhibits β-adrenergic signaling and lipolysis machinery in adipocytes

(A and B) *Ythdf2^{fl/fl}* and *Ythdf2^{Δadipo}* males were fed HFD for 16 weeks (A) or 8 weeks (B), and iWAT and eWAT were isolated. (A) WAT extracts were immunoblotted with the indicated antibodies. ATGL, CGI-58, HSL, and *Adrb3* levels were normalized to HSP90 (loading control) (*n* = 6 mice per group). (B) WAT was stimulated with 1 μM isoproterenol for 15 min *ex vivo*, and WAT extracts were immunoblotted with the indicated antibodies. HSL phosphorylation was normalized to HSL levels (*n* = 3 mice per group). PKA substrate phosphorylation was normalized to tubulin levels (*n* = 3 mice per group).

(C) C57BL/6J males were bilaterally injected with AAV-CAG-GFP and AAV-CAG-YTHDF2 vectors, respectively, and AAV-transduced iWAT was harvested 4 weeks later (Figure 3F). WAT extracts were immunoblotted with the indicated antibodies. ATGL, CGI-58, HSL, and *Adrb3* levels were normalized to p85 levels (loading control) (*n* = 6 mice per group).

(D) C3H preadipocytes differentiated into adipocytes and were then transduced with AAV8-CAG-GFP or AAV8-CAG-YTHDF2 vector. Cell extracts were immunoblotted with the indicated antibodies. ATGL, CGI-58, and HSL levels were normalized to HSP90 levels (loading control) (*n* = 3 per group).

Data are presented as mean ± SEM. **p* < 0.05 and ***p* < 0.01, two-sided unpaired *t* test (A) and (D).



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between *Ythdf2^{fl/fl}* and *Ythdf2^{Δadipo}* mice at both temperatures (Figure S6A).

To determine whether elevated ambient temperature has the opposite effect, we placed *Ythdf2^{Δadipo}* and *Ythdf2^{fl/fl}* males at thermoneutrality (30°C) on an HFD for 12 weeks. Remarkably, body weight, iWAT and eWAT mass, insulin resistance, and glucose tolerance were indistinguishable between *Ythdf2^{Δadipo}* and *Ythdf2^{fl/fl}* mice (Figures 6I, 6J, 6K, 6L, and S6B). In addition, hepatic lipid droplet accumulation and TAG levels were comparable between genotypes (Figures 6M and 6N). These results suggest that HFD and environmental temperature converge on the adipose YTHDF2/m⁶A pathway to regulate WAT growth and functions.

YTHDF2 is not required for BAT development

We previously reported that adipocyte-specific deletion of *Mettl14* (*Mettl14^{Δadipo}*) impairs BAT development, resulting in BAT whitening.¹⁰ Consistently, rectal temperature declined significantly more in *Mettl14^{Δadipo}* than in *Mettl14^{fl/fl}* mice following acute cold exposure (Figure 7A). In stark contrast, rectal temperature was comparable between *Ythdf2^{Δadipo}* and *Ythdf2^{fl/fl}* males after cold exposure, regardless of chow or HFD feeding (Figure 7B). Moreover, core body temperature during both the dark and light phases was normal in *Ythdf2^{Δadipo}* mice (Figure 7B). BAT temperature, assessed using a temperature-sensitive camera, was indistinguishable between *Ythdf2^{Δadipo}* and *Ythdf2^{fl/fl}* mice (Figure 7D). Female *Ythdf2^{Δadipo}* mice also displayed normal cold tolerance (Figure 7C). The number and size of lipid droplets in BAT were comparable between *Ythdf2^{Δadipo}* and *Ythdf2^{fl/fl}* mice on both chow diet and HFD (Figures S7A and 7E). Surprisingly, both UCP1 mRNA and protein levels were elevated in the BAT of *Ythdf2^{Δadipo}* mice compared to *Ythdf2^{fl/fl}* controls (Figures 7F and 7G). In contrast, UCP1 in BAT was dramatically downregulated in *Mettl14^{Δadipo}* mice (Figures 7F–7G). We next housed *Ythdf2^{Δadipo}* and *Ythdf2^{fl/fl}* mice at 16°C, 22°C, or 30°C and maintained them on HFD for 16 weeks. Lipid droplets in BAT were dramatically smaller in *Ythdf2^{Δadipo}* than in *Ythdf2^{fl/fl}* mice at 16°C, whereas no significant differences were observed between the two groups at 30°C (Figure 7H). Consistently, UCP1 expression in BAT was significantly higher in *Ythdf2^{Δadipo}* than in *Ythdf2^{fl/fl}* mice at both 16°C and 22°C but not 30°C (Figure 7I).

UCP1 expression in iWAT was also significantly higher in *Ythdf2^{Δadipo}* mice at 16°C but not at 30°C (Figure 7I). These results suggest that other m⁶A readers, rather than YTHDF2, act downstream of m⁶A writers to promote BAT development and function.

To identify the putative m⁶A readers, C3H cells were transfected with plasmids encoding YTHDF1, YTHDF2, YTHDF3, or YTHDC1, differentiated into brown adipocytes, and then stimulated with isoproterenol. Overexpression of YTHDF3 and YTHDC1, but not YTHDF2, significantly increased *Ucp1* expression under both baseline and isoproterenol-stimulated conditions (Figures S7B and S7C). These results suggest that YTHDF3 and YTHDC1 promote brown adipogenesis and BAT function.

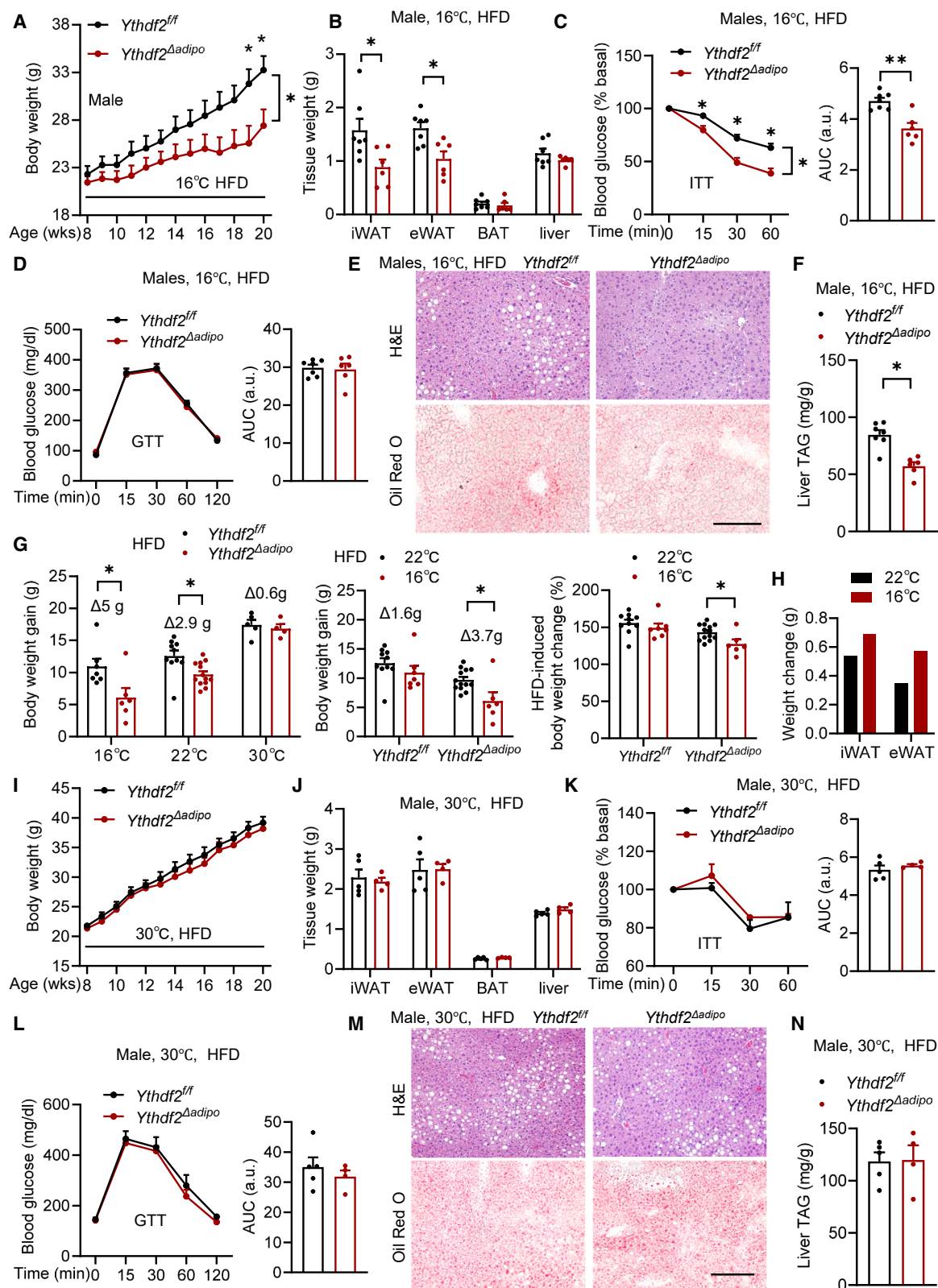
DISCUSSION

Adipose lipolysis mobilizes metabolic fuel to support life and is tightly regulated by the sympathetic nervous system and humoral mediators. ATGL catalyzes the rate-limiting step of lipolysis, and its activity is enhanced by cofactor CGI-58. The β-adrenergic pathway stimulates lipolysis by inducing phosphorylation and activation of ATGL and CGI-58. We and others previously reported that both diet-induced and genetic obesity are associated with adipose catecholamine resistance and lipolysis suppression.^{9,10} In this study, we identified the m⁶A reader YTHDF2 as a key regulator linking mRNA metabolism to catecholamine resistance and impaired lipolysis. Adipocyte-specific deletion of *Ythdf2* alleviated HFD-induced obesity, insulin resistance, and MASLD in *Ythdf2^{Δadipo}* mice. At the tissue and cellular levels, *Ythdf2* ablation increased WAT lipolysis *in vivo*, in WAT explants *ex vivo*, and in cultured adipocytes. Conversely, YTHDF2 overexpression suppressed iWAT lipolysis *in vivo* and in differentiated adipocytes *in vitro*. At the molecular level, we demonstrated that YTHDF2 directly binds to *Adrb3*, *Atgl*, and *Cgi-58* transcripts. This interaction was markedly attenuated by deletion of the m⁶A writer METTL14, suggesting that m⁶A modification is critical for YTHDF2 binding. Functionally, YTHDF2 promoted the decay of these transcripts. Pharmacological inhibition of METTL3 using STM2457 abrogated the ability of YTHDF2 to downregulate these mRNAs, further confirming the

Figure 5. HFD feeding stimulates YTHDF2-mediated degradation of m⁶A-marked *Adrb3*, *ATGL*, and *CGI-58* mRNA

(A) YTHDF2-bound mRNA was measured by YTHDF2-RIP and normalized to inputs in eWAT at 8 weeks of age on chow diet (*n* = 3 mice per group). a.u., arbitrary units. (B) C3H adipocytes were transduced with AAV8-CAG-FLAG-YTHDF2 or AAV8-CAG-GFP vector. YTHDF2-bound mRNA was measured by FLAG-RIP and normalized to inputs (*n* = 3 per group). (C) YTHDF2-bound mRNA was measured by YTHDF2-RIP and normalized to inputs in eWAT at 8 weeks of age on chow diet (*n* = 6 mice per group). (D and E) Gene expression in iWAT and eWAT on HFD for 16 weeks (normalized to 36B4 expression, *n* = 6 mice per group). (F) C57BL/6J males (8 weeks) were injected with AAV8-CAG-GFP or AAV8-CAG-YTHDF2 vector into iWAT as in Figure 3F. Gene expression was measured in iWAT 4 weeks later (*n* = 6 mice per group). (G–I) C3H preadipocytes were transfected with either GFP or YTHDF2 plasmids and differentiated into adipocytes. (G) Gene expression (normalized to 36B4 levels, *n* = 6 per group). (H) C3H adipocytes were treated with 1 μM actinomycin D for 0–6 h. mRNA was measured by qPCR and normalized to initial values (*n* = 3 per group). a.u., arbitrary units. (I) C3H adipocytes were treated with STM2457 (STM, 5 μM) for 4 h. Cell extracts were immunoblotted with the indicated antibodies. (J–L) C57BL/6J males (8 weeks) were fed HFD for 8 weeks, and iWAT and eWAT were harvested. (J) WAT extracts were immunoblotted with anti-YTHDF2 or anti-HSP90 antibodies. (K) WAT YTHDF2 levels were normalized to HSP90 expression (*n* = 8 mice per group). (L) YTHDF2-bound mRNA was measured by YTHDF2-RIP and normalized to inputs in WAT (*n* = 6 mice per group).

Data are presented as mean ± SEM. **p* < 0.05 and ***p* < 0.01, two-way ANOVA (H), and two-sided unpaired t test (A–G), (K), and (L). See also Figures S4 and S5.



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critical role of m⁶A modification. Consistent with these findings, YTHDF2 has been reported to promote degradation of other m⁶A-modified mRNAs.^{31,36} Importantly, *Adrb3*, *ATGL*, and *CGI-58* protein levels were elevated in the iWAT and eWAT of *Ythdf2*^{Δ*adipo*} mice, while YTHDF2 overexpression reduced their levels in iWAT. Furthermore, YTHDF2 deficiency enhanced adipose β3 receptor signaling *in vivo* and in WAT explants *ex vivo*. Together, these findings unveil the previously unrecognized YTHDF2/m⁶A-mediated pathways governing β3-adrenergic signaling and *ATGL/CGI-58*-mediated lipolysis. Disruption of these pathways promotes catecholamine resistance and suppressed lipolysis in obesity.

We observed that HFD feeding upregulates *METTL3*, *METTL14*, and YTHDF2 in WAT. TNFα, upregulated in obesity, directly enhanced YTHDF2 expression in adipocytes. We previously reported that obesity is associated with increased m⁶A modification on *Adrb3*, *Atgl*, and *Cgi-58* mRNAs.¹⁰ Consistently, we found that HFD feeding increased the binding of these transcripts to YTHDF2 in WAT. Based on these results, we propose a m⁶A-centric paradigm regulating lipolysis (Figure S7D). Obesogenic factors enhance the expression of *METTL3*, *METTL14*, and YTHDF2. The *METTL3/METTL14* complex installs m⁶A on mRNAs encoding key components of β-adrenergic signaling and the lipolysis machinery. YTHDF2 recognizes these m⁶A-marked transcripts and promotes their degradation, thereby acting as a molecular brake that dampens sympathetic signaling and lipolysis. Under obesogenic conditions, this YTHDF2-mediated brake is activated, contributing to catecholamine resistance, lipolysis suppression, WAT expansion, and metabolic dysfunction.

We and others previously reported that adipocyte-specific deletion of either *Mettl14* or *Mettl3* disrupts BAT development and function.^{10,39,40} Unexpectedly, deletion of adipose *Ythdf2* did not impair BAT development. *Ythdf2*^{Δ*adipo*} mice maintained normal adaptive thermogenesis in response to cold exposure. In brown adipocyte culture, overexpression of YTHDF3 or YTHDC1, but not YTHDF2, enhanced UCP1 expression. These observations suggest that distinct sets of m⁶A readers guide the development and function of WAT and BAT, respectively. During the revision of this manuscript, a study was published showing that deletion of *Ythdc1* in UCP1-expressing cells disrupts BAT development,⁴¹ confirming the critical role of YTHDC1 in brown adipogenesis. Surprisingly, UCP1 expression was elevated in the BAT of *Ythdf2*^{Δ*adipo*} mice under chronic cold

acclimation or HFD feeding conditions, suggesting that intrinsic YTHDF2 acts as a negative regulator of the adipose thermogenic program. Alternatively, increased lipolysis in *Ythdf2*^{Δ*adipo*} mice may lead to the generation of bioactive lipids that stimulate thermogenesis. Lipolysis derivatives act as the ligands for nuclear receptors PPARγ, PPARα, and PPARδ, which promote mitochondria biogenesis and/or activation, thus conferring metabolic benefits.^{42–44} Moreover, some of these lipid molecules function as lipokines that recruit and activate BAT and beige adipocytes.^{5,7,45–48} In addition, enhanced β-adrenergic signaling, which is known to activate BAT and beige fat,^{5,49} may further contribute to the increased adipose thermogenesis in *Ythdf2*^{Δ*adipo*} mice. It is likely that this enhanced adipose thermogenesis facilitates the clearance of circulating FFAs released from WAT, thereby alleviating MASLD observed in *Ythdf2*^{Δ*adipo*} mice.

Limitations of the study

Despite the importance of RNA m⁶A modification, our data cannot rule out the possibility that YTHDF2 suppresses lipolysis through a putative m⁶A-independent mechanism. Because *Ythdc1* was deleted embryonically, developmental alterations may also contribute to the observed phenotypes. Beyond *Adrb3*, *Atgl*, and *Cgi-58*, YTHDF2 may target additional transcripts involved in adipose function, which remain to be identified. Moreover, ambient temperature profoundly influences adipose YTHDF2 activity, but the underlying mechanism is unclear. Finally, the molecular processes by which YTHDF2 promotes the degradation of *Adrb3*, *Atgl*, and *Cgi-58* transcripts remain to be elucidated. Future studies will be required to address these outstanding questions.

RESOURCE AVAILABILITY

Lead contact

Requests for further information, resources, and reagents should be directed to and will be fulfilled by the lead contact, Dr. Liangyou Rui (rui@umich.edu).

Materials availability

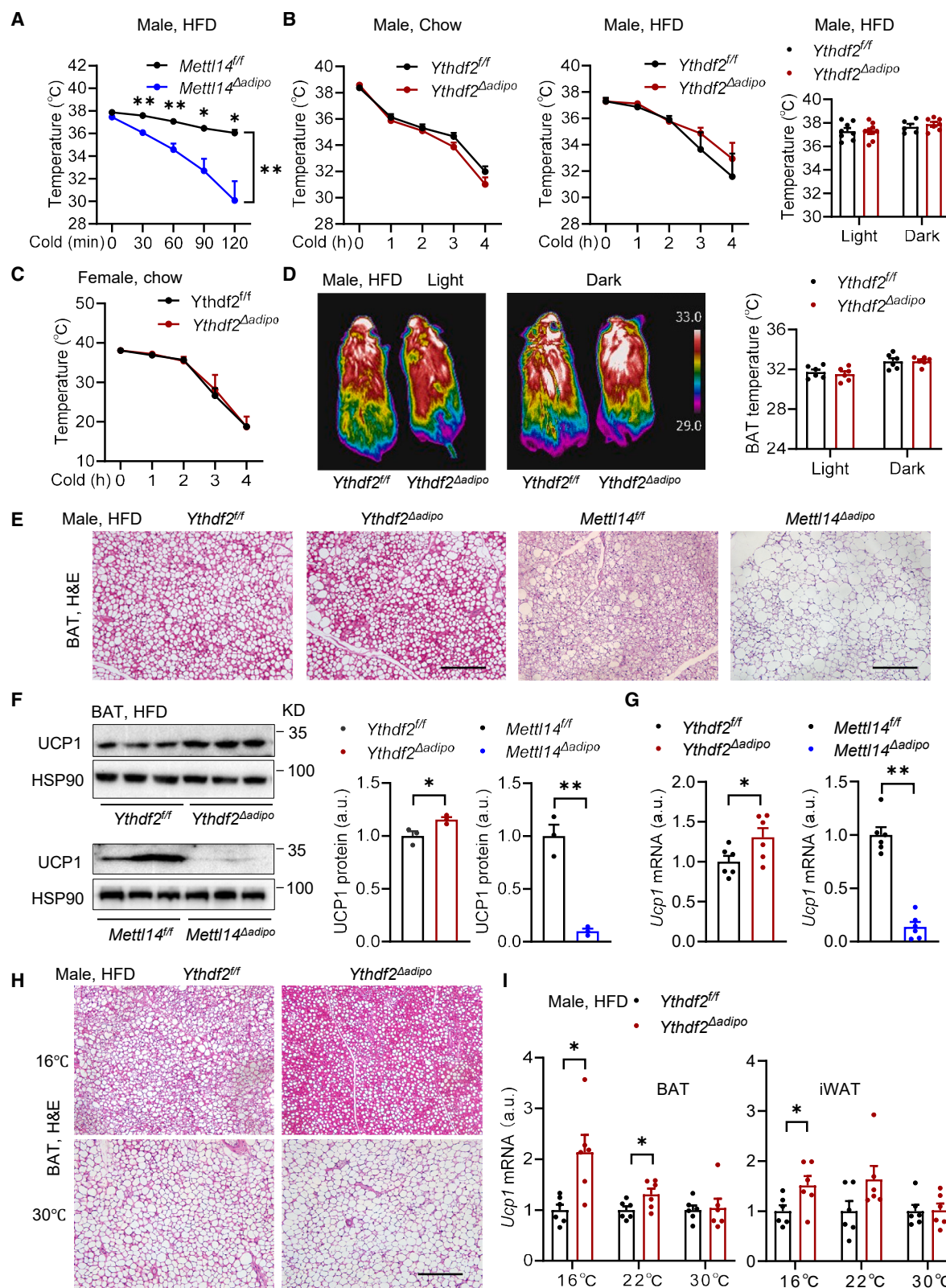
The study did not generate new reagents.

Data and code availability

- Original western blot images and microscopy data reported in this paper will be shared by the lead contact upon request. The authors declare that the data supporting the findings of this study are available within the paper and supplemental information files.

Figure 6. Ambient temperature shapes adipose YTHDF2 actions

(A–F) *Ythdf2*^{fl/fl} (n = 7) *Ythdf2*^{Δ*adipo*} (n = 6) male mice (8 weeks) were housed at 16°C and fed HFD. (A) Body weight. (B) Tissue weight after HFD for 16 weeks. (C) and (D) ITT and GTT on HFD for 12 weeks. AUC, area under the curve. (E) H&E and Oil red O staining of liver sections on HFD for 16 weeks. Scale bars, 200 μm. (F) Liver TAG after HFD for 16 weeks (normalized to liver weight). (G and H) *Ythdf2*^{fl/fl} and *Ythdf2*^{Δ*adipo*} males were fed HFD for 16 weeks at 16°C, 22°C, or 30°C ambient temperatures. (G) HFD-induced body weight gain was calculated. Differences in body weight gain (Δ) between *Ythdf2*^{fl/fl} and *Ythdf2*^{Δ*adipo*} mice were further calculated to be 5, 2.9, and 0.6 g for 16°C, 22°C, and 30°C, respectively. HFD-induced body weight changes were also presented as percentages of initial values. 16°C: *Ythdf2*^{fl/fl}, n = 7, *Ythdf2*^{Δ*adipo*}, n = 6; 22°C: *Ythdf2*^{fl/fl}, n = 10, *Ythdf2*^{Δ*adipo*}, n = 13; and 30°C: *Ythdf2*^{fl/fl}, n = 5, *Ythdf2*^{Δ*adipo*}, n = 4. (H) WAT difference between *Ythdf2*^{fl/fl} (average of 16°C: seven mice and 22°C: 10 mice) and *Ythdf2*^{Δ*adipo*} males (average of 16°C: six mice and 22°C: 13 mice) after HFD for 12 weeks at 22°C or 16°C. (I–N) *Ythdf2*^{fl/fl} (n = 5) and *Ythdf2*^{Δ*adipo*} males (n = 4) were housed at thermoneutrality (30°C) at 8 weeks of age and fed HFD. (I) Body weight. (J) Tissue weight after HFD for 16 weeks. (K) and (L) ITT and GTT after HFD for 12 weeks. (M) H&E and Oil red O staining of liver sections on HFD for 16 weeks. Scale bars, 200 μm. (N) Liver TAG after HFD for 16 weeks (normalized to liver weight). Data are presented as mean ± SEM. *p < 0.05 and **p < 0.01, two-way ANOVA (A), (C), (D), (I), (K), and (L), and two-sided unpaired t test (B–D), (F), (G), (J–L), and (N). See also Figure S6.



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- This paper does not report original code.
- Any additional information required to reanalyze the data reported in this paper is available from the [lead contact](#) upon reasonable request.

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AUTHOR CONTRIBUTIONS

R.Z., L.J., Q.K., and Q.Z. conducted the experiments; R.Z., L.J., Q.K., and L.R. analyzed the data; D.R. provided *Mettl14^{fl/fl}* and *Ythdf2^{fl/fl}* mice; R.Z., L.J., and L.R. designed the experiments and wrote the paper; and R.Z., L.J., Q.K., Q.Z., D.R., and L.R. edited the paper.

DECLARATION OF INTERESTS

The authors declare no competing interests.

STAR★METHODS

Detailed methods are provided in the online version of this paper and include the following:

- **KEY RESOURCES TABLE**
- **EXPERIMENTAL MODEL AND STUDY PARTICIPANT DETAILS**
 - Animals
 - Ethics statements
 - MEF preparations
 - C3H/10T1/2 cells and adipogenesis
 - AAV-CAG-YHRDF2 plasmids
- **METHOD DETAILS**
 - YTHDF2 overexpression in iWAT
 - GTT, ITT, and fat content
 - Cold tolerance tests
 - Infrared thermography measurements
 - Oil Red O staining and liver TAG
 - BODIPY staining
 - Transfection and brown adipogenesis
 - Lipolysis
 - RNA immunoprecipitation (RIP)
 - RNA decay assay
 - Immunoblotting
 - qPCR

QUANTIFICATION AND STATISTICAL ANALYSIS

SUPPLEMENTAL INFORMATION

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Figure 7. METTL14 but not YTHDF2 is essential for BAT development and thermogenesis

(A) *Mettl14^{fl/fl}* ($n = 6$) and *Mettl14^{Δadipo}* males ($n = 4$) were fed HFD for 14 weeks and subjected to cold tolerance tests (4°C). (B) Cold tolerance tests were performed on *Ythdf2^{fl/fl}* ($n = 6$) and *Ythdf2^{Δadipo}* ($n = 7$) males at 16 weeks of age on chow diet, or on *Ythdf2^{fl/fl}* ($n = 8$) and *Ythdf2^{Δadipo}* ($n = 9$) males on HFD for 12 weeks. In HFD-fed mice, core body temperature was also monitored in light (*Ythdf2^{fl/fl}*: $n = 8$ and *Ythdf2^{Δadipo}*: $n = 9$) and dark phases (*Ythdf2^{fl/fl}*: $n = 5$ and *Ythdf2^{Δadipo}*: $n = 7$). (C) Cold tolerance tests on female mice at 16 weeks of age on chow diet ($n = 5$ mice per group). (D) BAT temperature in *Ythdf2^{fl/fl}* ($n = 6$) and *Ythdf2^{Δadipo}* males ($n = 6$) on HFD for 12 wks. (E–G) *Ythdf2^{fl/fl}*, *Ythdf2^{Δadipo}*, *Mettl14^{fl/fl}*, and *Mettl14^{Δfat}* males were fed HFD for 16 weeks. (E) H&E staining of BAT sections. Scale bars, 200 μm . (F) BAT extracts were immunoblotted with anti-UCP1 and anti-HSP90 antibodies. UCP1 levels were normalized to HSP90 levels ($n = 3$ mice per group). (G) BAT gene expression after HFD for 16 weeks (normalized to 36B4 expression). *Ythdf2^{fl/fl}*: $n = 6$; *Ythdf2^{Δadipo}*: $n = 6$; *Mettl14^{fl/fl}*: $n = 6$; and *Mettl14^{Δadipo}*: $n = 6$. (H and I) Male mice were housed at 16°C , 22°C , or 30°C ambient temperature and fed HFD for 16 weeks. (H) H&E staining of BAT sections. Scale bars, 200 μm . (I) *Ucp1* mRNA levels in BAT and iWAT (normalized to 36B4 expression). 16°C : *Ythdf2^{fl/fl}*: $n = 6$; *Ythdf2^{Δadipo}*: $n = 6$; 22°C : *Ythdf2^{fl/fl}*: $n = 6$, *Ythdf2^{Δadipo}*: $n = 6$; and 30°C : *Ythdf2^{fl/fl}*: $n = 6$, *Ythdf2^{Δadipo}*: $n = 6$. Data are presented as mean $\pm$ SEM. * $p < 0.05$ and ** $p < 0.01$, two-way ANOVA test (A–C), and two-sided unpaired t tests (B), (D), (F), (G), and (I). See also Figure S7.

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STAR★METHODS

KEY RESOURCES TABLE

REAGENT or RESOURCE	SOURCE	IDENTIFIER
Antibodies		
YTHDF2 Rabbit polyAb	Cell Signaling Technology	Cat# 80014, RRID:AB_2923056
ATGL Rabbit mAb	Cell Signaling Technology	Cat# 2439, RRID:AB_2167953
CGI-58 (ABHD5) Rabbit polyAb	Proteintech	Cat# 12201-1-AP, RRID:AB_2220710
HSL Rabbit mAb	Cell Signaling Technology	Cat# 18381, RRID:AB_2798800
Adrb3 (β 3-AR) mouse mAb	Santa Cruz	Cat# sc-515763, RRID:AB_2868520
UCP1 Rabbit mAb	Cell Signaling Technology	Cat# 14670, RRID:AB_2687530
UCP1 Rabbit polyAb	Proteintech	Cat# 23673-1-AP, RRID:AB_2828003
PLIN1 (Perilipin) Rabbit mAb	Cell Signaling Technology	Cat# 9349, RRID:AB_10829911
pSer473 (Phospho-Akt (Ser473)) Rabbit mAb	Cell Signaling Technology	Cat# 4060, RRID:AB_2315049
pThr308 (Phospho-Akt (Thr308)) Rabbit mAb	Cell Signaling Technology	Cat# 2965, RRID:AB_2255933
Akt (pan) Rabbit mAb	Cell Signaling Technology	Cat# 4691, RRID:AB_915783
p-PAK sub (Phospho-PKA Substrate) Rabbit mAb	Cell Signaling Technology	Cat# 9624, RRID:AB_331817
pSer563 (Phospho-HSL (Ser563)) Rabbit polyAb	Cell Signaling Technology	Cat# 4139, RRID:AB_2135495
pSer660 (Phospho-HSL (Ser660)) Rabbit polyAb	Cell Signaling Technology	Cat# 4126, RRID:AB_490997
YTHDF1 Rabbit polyAb	Abclonal	Cat# A23773, RRID: N/A
YTHDF3 Rabbit polyAb	Abclonal	Cat# A8395, RRID: AB_2772933
YTHDC1 Rabbit polyAb	Abclonal	Cat# A22992, RRID: N/A
HSP90 Rabbit mAb	Cell Signaling Technology	Cat# 4877, RRID:AB_2233307
P85 (PI3 Kinase p85) Rabbit polyAb	Cell Signaling Technology	Cat# 4292, RRID:AB_329869
Tubulin (alpha-Tubulin) Rabbit polyAb	Cell Signaling Technology	Cat# 2144, RRID:AB_2210548
Bacterial and virus strains		
AAV8-CAG-Flag-YTHDF2	This paper	N/A
AAV8-CAG-GFP	This paper	N/A
Chemicals, peptides, and recombinant proteins		
actinomycin D	Sigma Aldrich	A9415-2
isoproterenol	Millipore Sigma	42-035-5100
TNF α	PeptoTech	300-01A
Palmitic acid	Sigma Aldrich	P0500
STM2457	DC Chemicals	DC53045
Critical commercial assays		
free glycerol reagent	Sigma	F-6428
NEFA-HR reagents	Wako Chemicals	999-34691
BODIPY TM 493/503	Invitrogen	D3922
Mycoplasma PCR Detection Kit	ABM	G238
Lipofectamine TM 3000 Transfection Reagent	Invitrogen	L3000001
Experimental models: Cell lines		
C3H/10T1/2	University of Michigan	N/A
Experimental models: Organisms/strains		
Mice wild type (C57/BL6)	Jackson Lab	N/A
<i>Mett14</i> ^{fl/fl} (C57/BL6)	Liu J et al. ⁵⁰	N/A
<i>Ythdf2</i> ^{fl/fl} (C57/BL6)	Li M et al. ³⁵	N/A

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REAGENT or RESOURCE	SOURCE	IDENTIFIER
Oligonucleotides		
Primers used for Real-time PCR reactions	This paper	See Table S1. Primer sequences used for qPCR analysis
Recombinant DNA		
pAAV-CAG-Flag-YTHDF2	This paper	N/A
Software and algorithms		
ImageJ2	NIH	https://imagej.net/downloads
GraphPad Prism10	GraphPad Software	https://www.graphpad.com/features
Biorender	Biorender website	https://BioRender.com/njsue17
Adobe Illustrator	Adobe	https://www.adobe.com/products/illustrator.html

EXPERIMENTAL MODEL AND STUDY PARTICIPANT DETAILS

Animals

Mett14^{flf} and *Ythdf2^{flf}* mice (C57BL/6J background) were described previously.^{35,50} *Mett14^{Δadipo}* mice were generated by crossing *Mett14^{flf}* mice with *Adiponectin-Cre* drivers.¹⁰ *Ythdf2^{Δadipo}* mice were generated by crossing *Ythdf2^{flf}* mice with *Adiponectin-Cre* mice. Mice were housed on a 12 h light-dark cycle at 22°C–24°C (regulator room temperature), 16°C, or 30°C, as indicated. They were fed *ad libitum* a chow diet (9% fat; TestDiet, St. Louis, MO) or HFD (60% fat in calories; Research Diets, New Brunswick, NJ). Both male and female mice were examined to assess potential influence of sex on phenotypes.

Ethics statements

Animal research was complied with relevant ethic regulations. Animal experiments were conducted following the protocols approved by the University of Michigan Institutional Animal Care and Use Committee (IACUC).

MEF preparations

Embryos (E17.5) were isolated and dissected in cold PBS. Internal organs and heads were removed, and the remaining tissues were pooled from 3 to 4 embryos (the same genotype) and minced in 1 mL trypsin buffer using a razor blade and incubated at 37°C for 30 min. Digestion was stopped by adding 4 mL DMEM supplemented with 10% FBS and P/S. MEF cells were dissociated by pipetting 10–20 times and grown in DMEM supplemented with 10% FBS.

C3H/10T1/2 cells and adipogenesis

Murine C3H cells (not authenticated) were grown at 5% CO₂ and 37°C in DMEM containing 25 mM glucose, 10% FBS, and P/S. Confluent C3H and MEF cells were cultured for additional 2 days and placed in an white adipogenesis cocktail for 4 days: DMEM containing 25 mM glucose, 10% FBS, 0.1 μM insulin, 1 μM dexamethasone (Sigma, D-1756), 0.5 mM 3-isobutyl-1-methylxanthine (Sigma, I-5879), 0.125 μM indomethacin (Sigma, I-7378), and 0.1 μM rosiglitazone (Cayman Chemical Company, Ann Arbor, Michigan). The cells were then grown for additional 2 days in DMEM supplemented with 10% FBS and 0.1 μM insulin and maintained in DMEM supplemented with 10% FBS.

AAV-CAG-YHRDF2 plasmids

Human *YTHDF2* cDNA was inserted into AAV-CAG plasmids using an In-Fusion kit (Takar Bio, Cat#638951).

METHOD DETAILS

YTHDF2 overexpression in iWAT

C57B6/J male mice (8 wks old) were anesthetized using isoflurane (VetOne, 3–4% for induction and 2% for maintenance). Hair was removed from the surgical site, and the surgical sites were cleaned with three alternating scrubs of 70% ethanol and chlorhexidine. Body temperature was maintained using a temperature-controlled surgical platform. Two 1–2 cm incisions were made near the lower back groin to expose iWAT. AAV8-CAG-YTHDF2 vector was injected into an iWAT depot (4–5 points, 5 μL per point), while AAV8-CAG-YTHDF2 vector (control) was injected into the contralateral iWAT depot. The incisions are closed using nylon sutures. The animals were placed in a warm cage and monitored until they were fully recovered.

GTT, ITT, and fat content

For GTT, mice were fasted overnight and intraperitoneally injected with glucose (1.5 g/kg); for ITT, mice were fasted for 6 h and intraperitoneally injected with insulin (0.6 units/kg). Tail vein blood glucose was measured after glucose or insulin injection. Fat content and lean body mass were measured using a dual-energy X-ray absorptiometry pDexa (Norland Stratec).

Cold tolerance tests

A mouse was placed in a pre-cooled cage (with water but without food). The cage was transferred into a rodent environmental chamber at 4°C (RIS33SD, Innovative Solutions, Beverly Hills, MI). Rectal temperature was measured hourly.

Infrared thermography measurements

The Infrared camera E60bx (FLIR Systems Inc., Wilsonville, USA) was placed 1 m above a mouse back to measure BAT and surface temperature. Thermographic images were analyzed to assess BAT temperature using FLIR Thermal Studio 2.0.11.0.

Oil Red O staining and liver TAG

Frozen liver sections were prepared using a Leica cryostat (Leica Biosystems Nussloch GmbH, Nussloch, Germany) and stained with 1% Oil Red O. Liver samples were homogenized in 1% acetic acid and then extracted by chloroform:methanol (2:1). The organic phase was dried by evaporation and dissolved in isopropanol. TAG was completely hydrolyzed to release glycerol with ethanolic KOH. Glycerol was measured using a free glycerol reagent (Sigma, Free Glycerol Reagent F-6428). TAG levels were calculated and normalized to liver weight.

BODIPY staining

Adipocytes (in 12-well plate) were washed with PBS and incubated for 15 min at 37°C in 0.5 mL of 2 μ M BODIPY staining solution. Cells were washed twice with PBS and fixed in 4% PFA for 30 min at room temperature.

Transfection and brown adipogenesis

C3H preadipocytes (approximately 80% confluence) were transfected with YTHDF1, 2, 3 or YTHDC1 plasmids (1 μ g) using Lipofectamine 3000 (Thermo Fisher Scientific), following manufacturer's instructions. Transfected cells in full confluence were differentiated into brown adipocytes for 96 h using a differentiation cocktail: DMEM supplemented with 25 mM glucose, 10% FBS, 0.1 μ M insulin, 1 μ M dexamethasone (Sigma, D-1756), 0.5 mM 3-isobutyl-1-methylxanthine (IBMX; Sigma, I-5879), 0.125 μ M indomethacin (Sigma, I-7378), 0.1 μ M rosiglitazone (Cayman Chemical, Ann Arbor, MI), and 1 nM triiodothyronine (T3, Sigma, T2877). The medium was then replaced with maintenance DMEM supplemented with 25 mM glucose, 10% FBS, 0.1 μ M insulin, 0.1 μ M rosiglitazone, and 1 nM T3 and refreshed every 48 h. Brown adipocytes were stimulated for 4 h with or without 1 μ M isoproterenol (ISO) and then used for qPCR and Western blot analyses.

Lipolysis

To measure lipolysis *in vivo*, male mice (16 wks) were fasted for 4 h and intraperitoneally injected with isoproterenol (10 mg/kg). Tail vein blood FFA and glycerol levels were measured 0–3 h after treatment using Free Glycerol Reagent (Sigma, F-6428) and NEFA-HR reagents (Wako Chemicals, Richmond, VA), respectively. For *ex vivo* assays, iWAT and eWAT were isolated from male mice and minced into 1 mm blocks in cold PBS. WAT explants were incubated at 37°C for 2 h in KRH buffer (136 mM NaCl, 4.7 mM KCl, 1.25 mM MgSO₄, 1.25 mM CaCl₂, 20 mM HEPES, pH 7.4, and 2% BSA), and then stimulated with 1 μ M isoproterenol. FFA and glycerol in KRH buffer were measured using NEFA-HR reagents (Wako Chemicals, Richmond, VA) and Free Glycerol Reagents (Sigma, F6428), respectively. Likewise, MEF-differentiated adipocytes were incubated in KRH buffer for 3 h and stimulated with 1 μ M isoproterenol for additional 3 h. FFA and glycerol in KRH buffer were measured using NEFA-HR reagents and Free Glycerol Reagents, respectively.

RNA immunoprecipitation (RIP)

WAT or adipocyte culture were homogenized on ice, using a glass Dounce homogenizer, for 20 min in NETN buffer: 20 mM Tris-Cl, pH 8.0; 100 mM NaCl, 1 mM EDTA, 0.5% NP-40, 1 mM Na₃VO₄, 100 mM NaF, 10 mM Na₄P₂O₇, 1 mM benzamidine, 10 μ g/mL aprotinin, 10 μ g/mL leupeptin, 1 mM phenylmethylsulfonyl fluoride, and RNasin (Promega, N2511). Cell lysates were centrifugated for 10 min at 4°C. Supernatants were incubated at 4°C for 2 h with 1 μ L anti-YTHDF2 or Flag-beads and then with 20 μ L protein-A agarose beads for additional 1 h (with rotations). The beads were washed with the ETN buffer, and their bound RNA was extracted for qPCR, using Trizol reagent (ThermoFisher, 15596026).

RNA decay assay

Adipocyte cultures were treated with transcription inhibitor actinomycin D (1 μ M) for 0–6 h. Total RNA was extracted for measuring mRNA abundance by qPCR (normalized to 18S levels).

Immunoblotting

WAT and BAT were homogenized on ice, using a glass Dounce homogenizer, in a lysis buffer (50 mM Tris, pH 7.5, 1% Nonidet P-40, 1% SDS, 150 mM NaCl, 2 mM EGTA, 1 mM Na_3VO_4 , 100 mM NaF, 10 mM $\text{Na}_4\text{P}_2\text{O}_7$, 1 mM benzamidine, 10 $\mu\text{g}/\text{mL}$ aprotinin, 10 $\mu\text{g}/\text{mL}$ leupeptin, 1 mM phenylmethylsulfonyl fluoride). Adipocyte cultures and livers were homogenized in the L-RIPA buffer (50 mM Tris, pH 7.5, 1% Nonidet P-40, 1% SDS, 150 mM NaCl, 2 mM EGTA, 1 mM Na_3VO_4 , 100 mM NaF, 10 mM $\text{Na}_4\text{P}_2\text{O}_7$, 1 mM benzamidine, 10 $\mu\text{g}/\text{mL}$ aprotinin, 10 $\mu\text{g}/\text{mL}$ leupeptin, 1 mM phenylmethylsulfonyl fluoride) using TissueLyser II (QIAGEN, Valencia, CA). Proteins were separated by SDS-PAGE and immunoblotted with the indicated antibodies ([key resources table](#)).

qPCR

Total RNA was extracted using TRIzol reagents (Invitrogen life technologies, Carlsbad, CA). The first-strand cDNAs were synthesized using random primers and M-MLV reverse transcriptase (Promega, Madison, WI). qPCR was performed using Radiant SYBR Green 2X Lo-ROX qPCR Kits (Alkali Scientific, Pompano Beach, FL), a StepOnePlus RT PCR Systems (Life Technologies Corporation, NY, USA). Primers for qPCR were listed in [Table S1](#).

QUANTIFICATION AND STATISTICAL ANALYSIS

Data was presented as means $\pm$ SEM. Differences between two groups were analyzed by 2-tailed Student's *t* test. Comparisons between 2 or more groups with more than 1 variable were analyzed by two-way ANOVA test using GraphPad Prism 10. *p* value less than 0.05 was considered statistically significant. **p* < 0.05 and ***p* < 0.01.